JIANGSU RUDONG LINGYANG WIND POWER PROJECT

Document Prepared by Longyuan (Beijing) Carbon Asset Management Technology Co., Ltd.

Project Title	Jiangsu Rudong Lingyang Wind Power Project
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Monitoring Period	01-Jan-2013 to 05-Dec-2017
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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

Jiangsu Rudong Lingyang Wind Power Project (hereafter referred as the project) is a grid connected renewable energy project located in Jiangsu Province. The objective of the project is to generate electricity using state-of-the-art wind power generation technology and to sell into the State Power Grid. The project achieves CO_2 emission reduction by replacing electricity generated by fossil fuel fired power plant connected into East China Power Grid.

Totally 33 sets of 1500kW wind turbines have been installed, providing a total capacity of 49.5MW. The area has relative rich wind resource. It is estimated that the annual generation of the project is 106.7 GWh. As a result, 98,526 tonnes of CO_2 emission reduction is expected to be generated.

The project reduces greenhouse gas emissions by supplying zero-emission wind power to East China Power Grid, which replaces the same amount of electricity generated by fossil fuel fired power plants connected into East China Power Grid, and therefore, avoids the CO_2 emissions in generating the same amount of electricity provided by the fossil fuel fired power plants.

The baseline scenario of the project is the electricity supply of equal amount as the project from the East China Power Grid. The baseline scenario of the project is the same against the scenario prior to the start of the implementation of the project activity.

The project began to construct on 10-Jun-2007, the first wind turbine was put into operation on 06-Dec-2007 and the project started full operation on 12-Jan-2008.

During this monitoring period, the wind turbines are in good operation condition. The operation environment and status in this monitoring period hasn't been changed from the previous monitoring periods. The emission reduction achieved by the project in this monitoring period (01-Jan-2013 to 05-Dec-2017) is 375,674 tCO_2e.

1.2 Sectoral Scope and Project Type

Sectoral scope: 1 energy industries (Renewable sources).

Project type I: Renewable energy Projects. Grid connected wind power project.

The project is not a grouped project.

1.3 Project Proponent

Organization name	Jiangsu Longyuan Wind Power Generation Co., Ltd.
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1.4 Other Entities Involved in the Project

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Role in the Project	Project developer
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1.5 Project Start Date

06-Dec-2007 (It is the date when the first wind turbine was put into operation).

1.6 Project Crediting Period

As described in Annex 1 of MR version 01 dated 6-May-2009， only emission reductions achieved from 6-Dec-2007 to 11- Jan-2009 will be considered as VCUs. Thus the crediting period was from 06-Dec-2007 to 11-Jan - 2009. The project is registered under VCS Standard Version 2007.1 and completed validation before 19-March-2020. Thus, it remains eligible to apply the crediting period requirements under VCS Standard Version 3, which shall be a maximum of ten years and may be renewed at most twice.

The project is also a registered CDM project (Ref.2169) with a 7*3 renewable project crediting period, the VCS issuance is not eligible for beyond the end of those 21 years. Furthermore, given that the CDM crediting period started on 11-january-2009 and the end date of 21 years is 10-January-2030 . So, the first VCS crediting period was from 06-Dec-2007 to 05-Dec-2017.

1.7 Project Location

The project is located in Rudong County, Jiangsu Province of China, its geographical coordinates are north latitude 32°35′5.2″-32°36′40.3″ and east longitude 121°24′15.8″-121°29′28.6″, and its altitude is 2.0~5.0 m. The detailed location of the project is shown in Figure 1.

Figure1. Map of Project Location

1.8 Title and Reference of Methodology

Title: Approved consolidated baseline methodology ACM0002 (Version 06) "Consolidated methodology for grid-connected electricity generation from renewable sources" and approved consolidated monitoring methodology ACM0002 (Version 06) "Consolidated monitoring methodology for zero-emissions grid-connected electricity generation from renewable sources".

The methodology also refers to the approved versions of the following tools:

Tool for the demonstration and assessment of additionality (Version 04)

1.9 Participation under other GHG Programs

The GHG emission reductions of 102,215 tCO_2e generated from 06-Dec-2007 to 11-Jan-2009 by the project has been issued as VCUs under VCS program.

The project is a registered CDM project with reference No. 2169. Total GHG emission reductions of 373,775 tCO_2 generated from 12-Jan-2009 to 31-Dec-2012 by the project has been issued as CERs under CDM program, the detailed information can be found at https://cdm.unfccc.int/Projects/DB/TUEV-SUED1218638332.96/view.

All emission reductions during 01-Jan-2013 to 05-Dec-2017 have not and will not seek CDM CER issuance. Emission reductions during this monitoring period will only seek issuance under VCS program.

1.10 Other Forms of Credit

Emission Trading Programs and Other Binding Limits

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other CDM that includes GHG allowance trading.

Other Forms of Environmental Credit

The project has not sought or received another form of GHG-related environmental credit, including renewable energy certificates, during this monitoring period.

1.11 Sustainable Development Contributions

The Project is expected to produce various benefits to society as well as the environment. Therefore, the Project can significantly contribute to national as well as local sustainable development, including:

- Reduction in GHG emissions: Currently, most of the electricity in the region is generated through conventional fossil fuel based thermal power plants, according to the China Electricity Power Yearbook. Through combustion of fossil fuels, they emit a large volume of Greenhouse gasses (GHG) into the atmosphere, which has a negative effect on global society. Generation of GHG emission free electricity, such as wind, can displace electricity generated by these fossil fuel based thermal plants, therefore reducing GHG emissions. The project activity achieves 375,674 tCO_2e emission reduction in this monitoring period.

Thus, the project achieved **SDG 13 Climate Action**[1].

- Reduction of fossil fuel use: The Project reduces reliance on imported fossil fuels, which can contribute to increasing China's energy security, and also improve local air quality as it reduces the emissions of SO_2, and NOx associated with fossil fuel use. During this monitoring period, 406,839.41 MWh of electricity from renewable sources had been exported to the power grid.

  Thus, the project achieved **SDG 7 Affordable and Clean Energy**[2].

[1] https://sdgs.un.org/goals/goal13

[2] https://sdgs.un.org/goals/goal7

Table 1: Sustainable Development Contributions

Row number	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Current Project Contributions	Contributions Over Project Lifetime
1)	7.2	7.2.1 Renewable energy share in the total final energy consumption	Implemented activities to increase	During this monitoring period, 406,839.41 MWh of electricity from renewable sources has been exported to the power grid.	From the operation start date of this project activity to the end of this monitoring period 922,316.879 MWh of electricity from renewable sources has been exported to the power grid.
2)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Implemented activities to increase	During this monitoring period, the project has achieved GHG emission reductions of 375,674 tCO_2e.	From the operation start date of this project activity to the end of this monitoring period, the project has achieved GHG emission reductions of 851,664 tCO_2e.

2 SAFEGUARDS

2.1 No Net Harm

In accordance with relevant environmental laws and regulations, an environmental impact assessment (EIA) of the project was completed in 2004 and has been approved by the Environmental Protection Administration of Jiangsu Province on 01-Dec-2004.

The project is likely to cause the following environmental impacts:

Main Potential Environmental Impacts Associated with the project

- Impacts from the construction of the wind farm include construction noise, dust as well as water and soil loss etc;
- Impacts from noise and the electromagnetism pollutions of the wind turbines during the operation period.
- Impacts on native vegetation and environment as a result of construction activities for windmill towers, transformers and access roads.
- Impacts on socio-economy from the construction and operation of the project.

Impacts on Air Environment

Wind Power plants are known to contribute to zero atmospheric pollution as no fuel combustion is involved during any stage of the operation. However, the sources of air pollution are mainly due to the construction activities including the transportation of construction material, road construction and improvement and cadre construction etc. The impacts on air environment are temporal that will be ended when the construction is completed. It is suggested that several measures shall be taken into account, such as prohibiting the construction under strong wind weather, reducing as much as possible the area of construction, spraying water as undertake construction, and reducing the speed of vehicles in the field. Hence, the air quality of the project site is relatively well, which meets the Class II of China Ambient Air Quality Standard (GB 3095-1996).

Impacts on Noise Environment

The noise of the project in construction phase is from vehicles and machines on-site. During the construction period of the project, the noise level was lower than 55dB at the distance to noise source of 250m, which meet the class I of China Environmental Noise Standard in Urban Area (GB3096-93). The project is located on seashore. Therefore, there was no noise disturbance issue. Based on the formula of declining of sound emitted from a non-directional source, it is estimated that the maximum noise effective distance of the project is 50m in daytime and 300m at night. Moreover, the magnitude of the impacts during the construction period will only exist temporarily and disappear with the completing of the construction period. Moreover, operational noise from the rotating blades is expected to be minimal due to the attenuation of noise beyond a distance of 300-500m from the wind turbines. The noise meets Class II of

China Environmental Noise Standard in Urban Area (GB3096-1993). At the same time, the closest residential area to the site of the project is over 5km away. Therefore, the noise of the project will not have impact on nearby residents.

Impacts on Water and Solid Waste

The windfarm does not consume any water, nor does it generate any wastewater in the operation phase. The possible negative impacts are the household wastewater and solid waste produced by builders and staff, and the waste earth from digging of the foundation in the construction phase. Under normal conditions with highly automated monitoring and control system, the household wastewater will be first treated in a septic tank, and then be disinfected to discharge for circumjacent virescence. Moreover, the amount of household solid waste will be very little, which will not have impact on the environment. Besides, the solid waste will be collected and moved to the landfill site of the nearest city. The waste earth from the digging should be firstly used for refilling. The rest of the waste earth should be placed in the low area of the site which should be replanted with grass. The quality of surface water meets the Class II of China Quality Standard for Sea Water (GB/T 14848-93). Following the suggestion, the water and solid waste should have no significant impact on the environment.

Impacts on telecommunications and television transmissions

Since set of 110kV substation will be constructed in the project, the electromagnetism impact of the project should be evaluated. Based on the analogies of the built windfarms, the result concludes that the operation of wind farm will not have electromagnetism impact on the nearby enterprises and residential areas that are 5 km away from the wind farms, and the electromagnetism impact is below the standard limit value specified in the Hygienic Standard on Power-frequency Electric Field Working Places (GB16203-1996). Therefore, the electromagnetism of this project in the operation phase doesn't impact the production and daily life of nearby enterprises and residents.

Impacts on Ecosystem Environment

A serious potential concern for wind farms is their impact on vegetation, animals and migrating birds. The minor quantity of soil erosion generated during the construction phase has no noticeable impact on soil use and the project proponent has made arrangements to dispose them in an environmentally acceptable manner. Moreover, there are no migratory birds / endangered species in the region of the project activity. After the completion of the construction, the surrounding area of permanent occupation of ground will be planted with vegetation; the temporary equipment will be taken away. The ground occupation will not have obvious impact on the local ecology impact. Therefore, the activities to be carried out will not generate any negative impact on the ecological environment.

Socio-Economic Impacts

The preliminary appraisal assumed a larger installed capacity and higher coal displacement in the project. The project is estimated to product GHG emission reduction through delivering electricity to the grid. So, the project generates eco-friendly, GHG free power that contributes to sustainable development of the region. Moreover, the locals have benefited economically through land sales and revenues. The project activity not only helps the uplift of skilled and unskilled manpower in the region, but also improves employment rate and livelihood of local populace in the vicinity of the project.

2.2 Local Stakeholder Consultation

2.2. Stakeholder consultation during registration stage under CDM scheme

1. Brief description how comments by local stakeholders have been invited and compiled:

On 21-Nov -2006, under the support of local government, the project owner successfully held a stakeholder meeting in Rudong County. Totally 14 stakeholder representatives participated the meeting, respectively from the Development and Reform Bureau of Rudong County, the Environmental Protection Bureau of Rudong County, the Management Council of Rudong Dayanggang Development Zone, and the Sanmin Village of Rudong County. (The registration form of the meeting is in the attachment.)

The comments in the meeting focus on following questions:

* Does the project benefit more to improve environment?

* What is the most important influence the project made?

* Does the project result in environmental problem?

* Did the migration happen? Did the project owner compensate for the land occupied by the project?

*Do you support the project?

2. Summary of the comments received:

Every stakeholder representative expressed the comments for the project. No opposite comment was received. The summary of the comments is as follows:

Comments from the local government: Wind power projects are environment friendly projects and are highly encouraged by China central government. Both the Development and Reform Commission of Rudong County and Environmental Protection Administration of Rudong County highly support the assessment of local wind resources and the development of wind power projects. They hope the successful implementation of the project can diversify local power mix, mitigate electricity shortage, and promote the development of local tourism and other tertiary industries.

Comments from villager representatives: The project site is located on seashore of Huang Sea, and there are no permanent residents and cropland nearby. Therefore, the construction of the wind power plant does not result in moving local residents or noise disturbance. The local villagers are satisfied with compensation by the project owner for occupation on part of their temporary aquafarm. The villager representatives are aquafarm owners from villages relatively near to the project site. They do not think that the project will have significant impact on their aquafarm or daily life.

3. Report on how due account was taken of any comments received:

Since there is no negative comment received, it's no need to make adjustment on design, construction and operation of the project. But the project owner and contractor still take measures to mitigate impact on environment during construction.

2.2.2 Stakeholder consultation during this monitoring period

Communications with local stakeholders are being carried out at periodic intervals. In this monitoring period, the project owner carried out questionnaire survey for the local stakeholder to collect the relevant comments and suggestions in Aug 2015. And there are no negative comments received for the project. In line with VCS requirements all the processes have been implemented to receive comments from local stakeholders as well as communicate with them at periodic intervals.

2.3 AFOLU-Specific Safeguards

Not applicable.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The project began to construct on 10-Jun-2007, and the project started full operation on 12-Jan-2008. The objective of the project is to generate electricity using state-of-the-art wind power generation technology and to sell into the State Power Grid. The project achieves CO_2 emission reduction by replacing electricity generated by fossil fuel fired power plant connected into East China Power Grid.

Totally 33 GE1.5 SE wind turbines with a nominal capacity of 1.5 MW, manufactured by *GE Corporation*, were installed, providing a total capacity of 49.5 MW and supplying an average annual generation of 106.7 GWh to the power grid. The main technical specifications of wind turbines are as follows:

Table 2 Main technical Characteristics of wind turbines for the project

Operating data	
Rated capacity:	1500kW
Cut-in wind speed:	4m/s
Cut-out wind speed (10 min. avg.):	25m/s

Rated wind speed:	13m/s
Wind Class - IEC:	Ib
Rotor	
Number of rotor blades:	3
Rotor diameter:	70.5m
Swept area:	3904m^2
Rotor speed(variable):	12.0-22.2rpm
Tower	
Hub heights-IEC:	54.7/64.7m
Power control	Active blade pitch control
Gearbox	Three step planetary spur gear system
Generator	Doubly fed, three-phase induction (asynchronous)
Converter	Pulse-width modulated IGBT frequency converter
Braking system (fail-safe)	Electromechanical pitch control for each blade (3 self-contained systems)
	Hydraulic parking brake
Yaw system	Electromechanical driven with wind direction sensor and automatic cable unwind
Control system	PLC (Programmable logic controller) with remote control and monitoring system
Noise reduction	Impact noise insulation of the gearbox and generator
	Sound reduced gearbox
	Noise reduced nacelle
	Rotor blades with minimized noise level
Lightning protection system	Lightning receptors installed along blades
	Surge protection in electrical components
Tower design	Multi-coated, conical tubular steel tower with safety ladder to the nacelle
	Load lifting system, load-bearing capacity over 200 kg

Each turbine has a 690V-to-35kV transformer, from which export side links into a joint unit. All these lines are combined into three joint units, and then be sent into the on-site 110kV transformer substation, finally link into the 220KV Wuyi substation. The wind turbines and transmission facility could be monitored and controlled either by onsite central control room or by control room of Rudong Electricity Supply Bureau and Nantong Power Corporation remotely.

During this monitoring period, the project had a smooth data transfer and good grid connection, and no special events that may impact the GHG emission reductions or removals and monitoring. No events or situations occurred during the monitoring period, which may impact the applicability of the methodology.

3.2 Deviations

3.2.1 Methodology Deviations

There is no methodology deviation for the project.

3.2.2 Project Description Deviations

As described in Annex1 of MR version 01 dated 6-May-2009 , only emission reductions achieved from 06-Dec-2007 to 11-Jan-2009 will be considered as VCUs. Thus, the crediting period was from 06-Dec-2007 to 11-Jan-2009.

A deviation is requested for the project crediting period. The first crediting period has been changed from 06-Dec-2007 - 11-Jan-2009 to 06-Dec-2007-05-Dec-2017 as the deviation.

As per Appendix 3 of VCS standard version 4.3, Registered projects and projects that complete validation on or before 19 March 2020 remain eligible to apply the crediting period requirements under VCS standard version 3. The project is registered under VCS Standard Version 2007.1 and completed validation before 19-Mar-2020. Thus, the crediting period requirements under VCS standard version 3 has been applied by the project.

As per section 3.8 of VCS Standard version 3. The project crediting period shall be a maximum of ten years and may be renewed at most twice for non-AFOLU projects. Projects registered under other GHG programs are not eligible for VCU issuance beyond the end of the total project crediting period under those programs. Renew validation report shall be issued after the end of the (previous) project crediting period but within two years after the end of the (previous) project crediting period. The project started operation on 06-Dec-2007. The end date of the CDM 3*7 crediting period of the project is 11-Jan-2030. Thus, the total VCS crediting period was determined from 06-Dec-2007 to 11-Jan-2030, of which the first VCS crediting period is from 06-Dec-2007~05-Dec-2017. Since the project did not conduct the validation of crediting period renewal within the two years after the end of the first VCS crediting period, the VCS crediting period of the project ends on 05-Dec-2017.

3.3 Grouped Projects

The Project is not a grouped project.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

The parameter that is available at validation of registered PDD, The parameter is used to calculate $EF_{,y}$.

Data / Parameter	EF_y
Data unit	tCO_2e/MWh
Description	Baseline emission factor: the combined emission factor of the project grid system.
Source of data	Registered PDD
Value applied	0.9234
Justification of choice of data or description of measurement methods and procedures applied	Fixed during the first crediting period.
Purpose of Data	Calculation of baseline emissions
Comments	The emission factor of the Project was ex-ante determined and is fixed during the first crediting period. All data and parameters had been determined at registration.

4.2 Data and Parameters Monitored

Data / Parameter	EG_y
Data unit	MWh/yr
Description	Net electricity delivered to the grid by the project
Source of data	Meter reading records
Description of measurement methods and procedures to be applied	The calculation of this parameter is shown as below: $EG_{,y}=EG_{export,y}- EG_{import,y}$
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	406,839.41
Monitoring equipment	Measured by Bi-directional meters (M1 as main meter and M2 as check meter) installed at the outlet of 35kv to 110kv transformer including the electricity supplied to the grid ($EG_{export,y}$) and

	imports from the grid ($EG_{import,y}$). (Meters information and calibration see Table 3 below). During this monitoring period, main meters M1 was running properly, hence meter readings from check meter M2 has not been applied.
QA/QC procedures to be applied	The metering equipment for the monitoring of $EG_{export,y}$ and $EG_{import,y}$ are calibrated annually for accuracy by a qualified third party in accordance with industry standards. Electricity exported to/imported from the grid were double checked by sales evidence, such as electricity transaction notes (ETN). The data have been archived electronically and kept during the creating period and 2 years after.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	-

Data / Parameter	$EG_{export,y}$
Data unit	MWh/yr
Description	Electricity generation supplied to the power grid by the project in year y
Source of data	Meter reading records
Description of measurement methods and procedures to be applied	Continuously measured and monthly recorded. Data are archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	408,448.77
Monitoring equipment	Measured by Bi-directional meters (M1 as main meter and M2 as check meter) installed at the outlet of 35kv to 110kv transformer. (Meters information and calibration see Table 3 below). During this monitoring period, main meters M1 was running properly, hence meter readings from check meter M2 has not been applied.

QA/QC procedures to be applied	The meters are calibrated annually for accuracy by a qualified third party in accordance with industry standards. Electricity exported to the grid were double checked by sales evidences, such as electricity transaction notes (ETN). The data have been archived electronically and kept during the creating period and 2 years after.
Purpose of data	Baseline emission calculation
Calculation method	N/A
Comments	-

Data / Parameter	$EG_{import,y}$
Data unit	MWh/yr
Description	Electricity imported from the power grid by the project in year y
Source of data	Meter reading records
Description of measurement methods and procedures to be applied	Continuously measured and monthly recorded. Data are archived for 2 years following the end of the crediting period by means of electronic and paper backup.
Frequency of monitoring/recording	Continuously measured and monthly recorded
Value applied	1,609.36
Monitoring equipment	Measured by Bi-directional meters (M1 as main meter and M2 as check meter) installed at the outlet of 35kv to 110kv transformer. (Meters information and calibration see table 3 below). During this monitoring period, main meters M1 was running properly, hence meter readings from check meter M2 has not been applied.
QA/QC procedures to be applied	The meters are calibrated annually for accuracy by a qualified third party in accordance with industry standards. Electricity imported from the grid were double checked by sales evidence, such as electricity transaction notes (ETN). The data have been archived electronically and kept during the creating period and 2 years after.
Purpose of data	Baseline emission calculation

Calculation method	N/A
Comments	-

4.3 Monitoring Plan

1. Data monitored

As the baseline emission factor is based on ex-ante calculation, the main data monitored is the electricity supplied to the grid by the project and the electricity imported from the power grid by the project. It is continuously measured and recorded by the project owner at 24:00 every day and then reports to the grid company monthly for confirmation purpose. The local grid company issues the ETNs monthly which is used to crosscheck the monthly recording data. According to the requirement of local grid company, the ETN of export to the grid and import from the grid is recorded on 24:00 of the last day of every month.

2. Monitoring organization

The Monitoring Plan clearly states the persons and the responsibilities of the QC team of the project; they are responsible for monitoring the electricity generation of the project. The assigned staff is in accordance with the registered PDD.

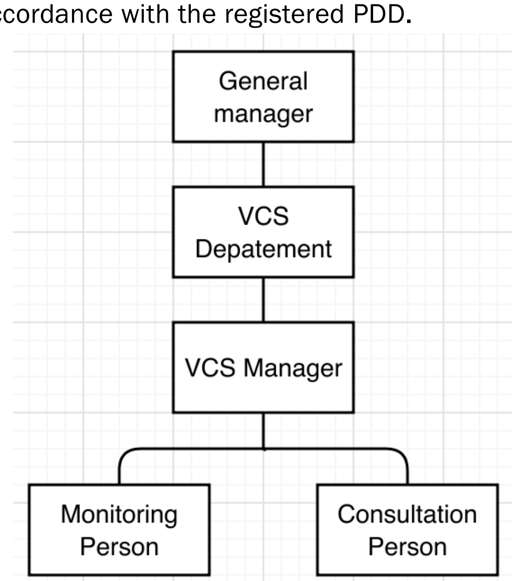

Figure 2 Monitoring organization

The Management Group is responsible for the VCS management of the project, which specifically includes selecting a VCS consultant company, providing support to the selected consultant company, assisting the selected consultant company in selecting VER buyer and determine VCU price, assisting the selected consultant company in applying for China DNA approval, providing support in the due diligence of the selected buyer, providing support in the VCS validation and registration by a selected VVB, and providing support in the verification by a selected VVB.

The Monitoring Person is responsible for reading and calibration of the meter, recording of the readings, and reporting of readings to local electric power company and/or VVB.

The Consultation Person is responsible for selecting a VCS consultant company and providing support to its work in developing PDD, selecting VER buyer, and applying for DNA approval.

3 Monitoring Devices

Two meters, main meter (M1) and check meter (M2), are installed in this project. They are both installed at the outlet of 35kv to 110kv transformer. All the meters are the multifunctional electricity meters (accuracy degree is 0.2S, bidirectional). The electricity supplied to the grid and the electricity imported from the grid are both measured according to the main meter. In order to measure electricity when the main meter is out of order, the check meter is for backup usage.

All the meters are the multifunctional electricity meters (accuracy degree is 0.2S, bidirectional). The electricity supplied to the grid and the electricity imported from the grid is measured according to the main meter. In order to measure electricity when the main meter is out of order, the check meter is for backup usage.

All the data and relevant documents are archived for 2 years following the end of the crediting period by means of electronic and paper backup.

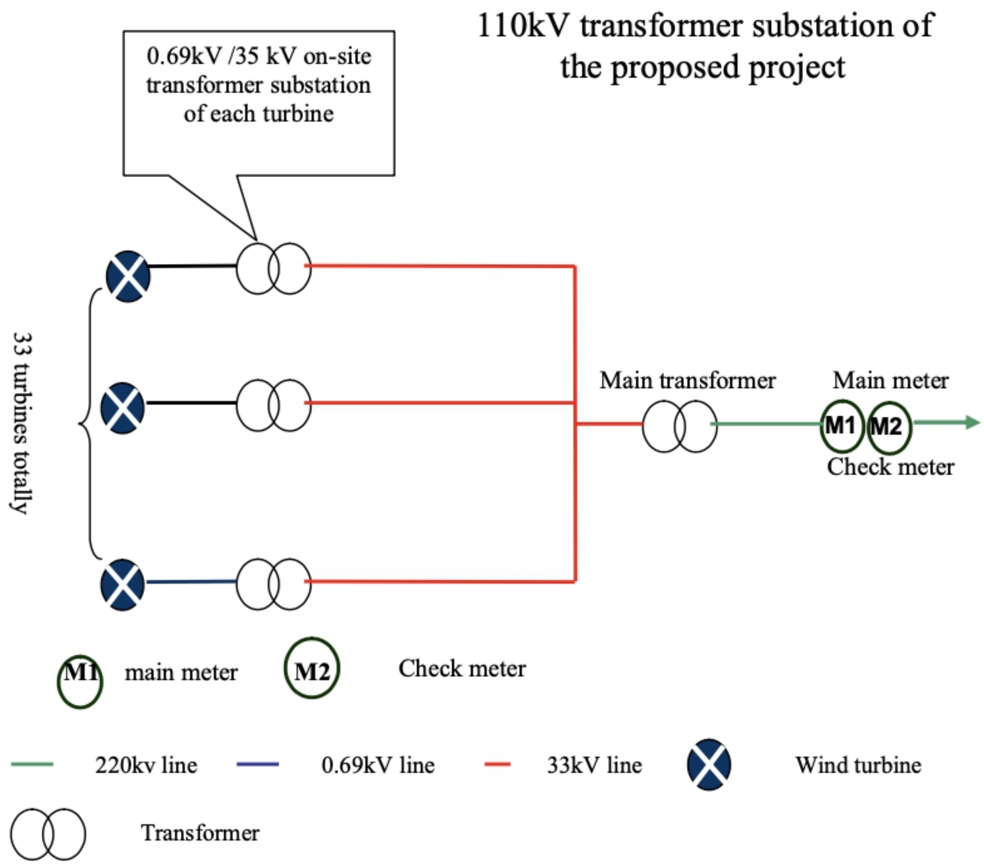

Figure 3 Main diagram of the project

4 Monitoring Practice and Handling of Emergency

Should reading of meter be inaccurate by more than the allowable error, or otherwise functioned improperly, the electricity supplied to the grid by the project shall be determined by:

1) First, by reading the check meter, unless a test by either party reveals it is inaccurate.

2) If main meter and check meter are inaccurate, by reading the self-carried meters of wind turbines, unless a test by either party reveals they are inaccurate.

3) if the self-carried meters of wind turbines are not with acceptable limits of accuracy or are otherwise performing improperly, the project owner and the local grid shall jointly prepare an estimate of the correct reading; and

4) If the project owner and the local grid fail to agree the estimate of the correct reading, then the matter can be referred for arbitration according to agreed procedures.

5 Data Management System

In order to facilitate verification, all data relevant to monitoring are carefully kept and stored by the project owner.

The following table below outlines the key documents relevant to monitoring and verification of the emission reductions of the project.

No.	Document	Source
F-1	PDD, including the electronic spreadsheets and supporting documentation	The project owner
F-2	The monitoring report on the net electricity supplied to the power grid	The project owner
F-3	Report on maintenance and calibration of metering equipment	The project owner

6 Meter Calibration

Main meter and check meter were calibrated during this monitoring period to make sure the accuracy. After calibration, two meters were sealed. The calibration records are available for verification. The calibrated accuracy level of the meters is 0.2s.

	Serial No. of the meter	Installation site	Calibration	Validity date
Main meter	85093984	at the outlet of the 35kV-to-110kV transformer	18- Apr -2012 10-Mar-2013 10-Feb-2014 10-Jan-2015 10-Dec-2015 10-Nov-2016 10-Oct-2017	17- Apr -2013 09-Mar-2014 09-Feb-2015 09-Jan-2016 09-Dec-2016 09-Nov-2017 09-Oct-2018
Check meter	85060035	at the outlet of the 35kV-to-110kV	18- Apr -2012	17- Apr -2013

		transformer	10-Mar-2013 10-Feb-2014 10-Jan-2015 10-Dec-2015 10-Nov-2016 10-Oct-2017	09-Mar-2014 09-Feb-2015 09-Jan-2016 09-Dec-2016 09-Nov-2017 09-Oct-2018

7 Training

The monitoring staff has all received sufficient training in terms of monitoring and verification. They have received general training on wind power project operation organized by the project owner, including reading and calibration of meters, recording of meter readings, and reporting of readings. They all have the professional qualifications of operation and maintenance in electricity power industry. Furthermore, they have received VCS training organized by the project owner.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

According to ACM0002, the baseline emission BE_y during the monitoring period results from:

$BE_y = EG_y \times EF_y$

$EG_y = EG_{export,\,y} - EG_{import,\,y}$

Where:

BE_y is the baseline emissions in year y (tCO_2/yr);

EG_y is the quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr);

EF_y is the combined margin baseline emission factor of the NWPG;

$EG_{export,\,y}$ is the Electricity exported to the grid by the Project.

$EG_{import,\,y}$ is the Electricity imported from the grid by the Project.

The net electricity each year supplied by the project activity to the grid, EG_y is listed in following Table 4.

Table 3 The calculation of EG_y

Period	Electricity exported to the grid by the project ($EG_{export,y}$) (MWh)			Electricity imported from the grid by the project ($EG_{import,y}$) (MWh)			Net electricity supplied to the grid EG_y (MWh)
	Values from main meter (M1) readings	Values from ETNs	Conservative	Values from main meter (M1) readings	Values from ETNs	Conservative	**G=C-F**
	A	B	**C=Min (A, B)**	D	E	**F=Max (D, E)**	
01-Jan-2013 to 31-Jan-2013	5,900.88	5,900.88	5,900.88	38.88	38.88	38.88	5,862.00
01-Feb-2013 to 28-Feb-2013	7,507.04	7,507.04	7,507.04	30.69	30.69	30.69	7,476.35
01-Mar-2013 to 31-Mar-2013	12,193.13	12,193.13	12,193.13	17.30	17.30	17.30	12,175.83
01-Apr-2013 to 30-Apr-2013	11,276.74	11,276.74	11,276.74	9.56	9.56	9.56	11,267.18
01-May-2013 to 31-May-2013	8,954.28	8,954.28	8,954.28	14.12	14.12	14.12	8,940.16
01-Jun-2013 to 30-Jun-2013	6,523.57	6,523.57	6,523.57	21.48	21.48	21.48	6,502.09
01-Jul-2013 to 31-Jul-2013	7,108.84	7,108.84	7,108.84	15.03	15.03	15.03	7,093.81
01-Aug-2013 to 31-Aug-2013	7,375.90	7,375.90	7,375.90	17.85	17.85	17.85	7,358.05
01-Sep-2013 to 30-Sep-2013	5,611.44	5,611.44	5,611.44	29.12	29.12	29.12	5,582.32
01-Oct-2013 to 31-Oct-2013	9,512.67	9,512.67	9,512.67	11.96	11.96	11.96	9,500.71
01-Nov-2013 to 30-Nov-2013	6,756.61	6,756.61	6,756.61	19.28	19.28	19.28	6,737.33
01-Dec-2013 to 31-Dec-2013	6,791.22	6,791.22	6,791.22	25.81	25.81	25.81	6,765.41
01-Jan-2013 to 31-Dec-2013	**95,512.32**	**95,512.32**	**95,512.32**	**251.08**	**251.08**	**251.08**	**95,261.24**
01-Jan-2014 to 31-Jan-2014	7,640.10	7,640.10	7,640.10	32.00	32.00	32.00	7,608.10
01-Feb-2014 to 28-Feb-2014	9,909.85	9,909.85	9,909.85	27.46	27.46	27.46	9,882.39
01-Mar-2014 to 31-Mar-2014	10,121.31	10,121.31	10,121.31	11.96	11.96	11.96	10,109.35
01-Apr-2014 to 30-Apr-2014	7,487.52	7,487.52	7,487.52	30.81	30.81	30.81	7,456.71
01-May-2014 to 31-May-2014	8,628.80	8,628.80	8,628.80	17.80	17.80	17.80	8,611.00
01-Jun-2014 to 30-Jun-2014	4,780.40	4,780.40	4,780.40	33.58	33.58	33.58	4,746.82
01-Jul-2014 to 31-Jul-2014	4,661.29	4,661.29	4,661.29	30.28	30.28	30.28	4,631.01
01-Aug-2014 to 31-Aug-2014	5,890.42	5,890.42	5,890.42	52.92	52.92	52.92	5,837.50

01-Sep-2014 to 30-Sep-2014	8,849.05	8,849.05	8,849.05	20.16	20.16	20.16	8,828.89
01-Oct-2014 to 31-Oct-2014	10,205.46	10,205.46	10,205.46	26.34	26.34	26.34	10,179.12
01-Nov-2014 to 30-Nov-2014	6,045.40	6,045.40	6,045.40	31.54	31.54	31.54	6,013.86
01-Dec-2014 to 31-Dec-2014	7,430.97	7,430.97	7,430.97	29.47	29.47	29.47	7,401.50
01-Jan-2014 to 31-Dec-2014	**91,650.57**	**91,650.57**	**91,650.57**	**344.32**	**344.32**	**344.32**	**91,306.25**
01-Jan-2015 to 31-Jan-2015	7,536.19	7,536.19	7,536.19	24.43	24.43	24.43	7,511.76
01-Feb-2015 to 28-Feb-2015	7,333.73	7,333.73	7,333.73	34.92	34.92	34.92	7,298.81
01-Mar-2015 to 31-Mar-2015	4,077.18	4,077.18	4,077.18	25.82	25.82	25.82	4,051.36
01-Apr-2015 to 30-Apr-2015	9,343.69	9,343.69	9,343.69	24.60	24.60	24.60	9,319.09
01-May-2015 to 31-May-2015	6,175.43	6,175.43	6,175.43	20.49	20.49	20.49	6,154.94
01-Jun-2015 to 30-Jun-2015	5,520.69	5,520.69	5,520.69	22.96	22.96	22.96	5,497.73
01-Jul-2015 to 31-Jul-2015	5,328.20	5,328.20	5,328.20	35.79	35.79	35.79	5,292.41
01-Aug-2015 to 31-Aug-2015	6,138.17	6,138.17	6,138.17	43.72	43.72	43.72	6,094.45
01-Sep-2015 to 30-Sep-2015	4,394.18	4,394.18	4,394.18	40.85	40.85	40.85	4,353.33
01-Oct-2015 to 31-Oct-2015	4,496.52	4,496.52	4,496.52	39.14	39.14	39.14	4,457.38
01-Nov-2015 to 30-Nov-2015	7,026.08	7,026.08	7,026.08	35.02	35.02	35.02	6,991.06
01-Dec-2015 to 31-Dec-2015	4,845.18	4,845.18	4,845.18	42.65	42.65	42.65	4,802.53
01-Jan-2015 to 31-Dec-2015	**72,215.24**	**72,215.24**	**72,215.24**	**390.39**	**390.39**	**390.39**	**71,824.85**
01-Jan-2016 to 31-Jan-2016	8,078.18	8,078.18	8,078.18	27.80	27.80	27.80	8,050.38
01-Feb-2016 to 29-Feb-2016	6,985.51	6,985.51	6,985.51	29.10	29.10	29.10	6,956.41
01-Mar-2016 to 31-Mar-2016	8,034.98	8,034.98	8,034.98	27.96	27.96	27.96	8,007.02
01-Apr-2016 to 30-Apr-2016	6,319.43	6,319.43	6,319.43	27.34	27.34	27.34	6,292.09
01-May-2016 to 31-May-2016	7,557.79	7,557.79	7,557.79	27.93	27.93	27.93	7,529.86
01-Jun-2016 to 30-Jun-2016	4,002.89	4,002.89	4,002.89	28.25	28.25	28.25	3,974.64
01-Jul-2016 to 31-Jul-2016	4,059.01	4,059.01	4,059.01	29.12	29.12	29.12	4,029.89
01-Aug-2016 to 31-Aug-2016	4,569.88	4,569.88	4,569.88	24.64	24.64	24.64	4,545.24
01-Sep-2016 to 30-Sep-2016	7,444.31	7,444.31	7,444.31	39.79	39.79	39.79	7,404.52
01-Oct-2016 to 31-Oct-2016	8,284.56	8,284.56	8,284.56	12.04	12.04	12.04	8,272.52

01-Nov-2016 to 30-Nov-2016	7,537.41	7,537.41	7,537.41	30.68	30.68	30.68	7,506.73
01-Dec-2016 to 31-Dec-2016	7,080.53	7,080.53	7,080.53	21.23	21.23	21.23	7,059.30
01-Jan-2016 to 31-Dec-2016	**79,954.48**	**79,954.48**	**79,954.48**	**325.88**	**325.88**	**325.88**	**79,628.60**
01-Jan-2017 to 31-Jan-2017	5,109.41	5,109.41	5,109.41	29.78	29.78	29.78	5,079.63
01-Feb-2017 to 28-Feb-2017	7,187.82	7,187.82	7,187.82	24.14	24.14	24.14	7,163.68
01-Mar-2017 to 31-Mar-2017	6,556.34	6,556.34	6,556.34	23.61	23.61	23.61	6,532.73
01-Apr-2017 to 30-Apr-2017	7,546.16	7,546.16	7,546.16	15.85	15.85	15.85	7,530.31
01-May-2017 to 31-May-2017	6,448.73	6,448.73	6,448.73	20.73	20.73	20.73	6,428.00
01-Jun-2017 to 30-Jun-2017	4,141.94	4,141.94	4,141.94	28.50	28.50	28.50	4,113.44
01-Jul-2017 to 31-Jul-2017	3,178.34	3,178.34	3,178.34	29.69	29.69	29.69	3,148.65
01-Aug-2017 to 31-Aug-2017	5,700.84	5,700.84	5,700.84	43.76	43.76	43.76	5,657.08
01-Sep-2017 to 30-Sep-2017	5,769.24	5,769.24	5,769.24	30.32	30.32	30.32	5,738.92
01-Oct-2017 to 31-Oct-2017	10,621.16	10,621.16	10,621.16	19.95	19.95	19.95	10,601.21
01-Nov-2017 to 30-Nov-2017	6,763.27	6,763.27	6,763.27	30.97	30.97	30.97	6,732.30
01-Dec-2017 to 05-Dec-2017	92.91	92.91	92.91	0.39	0.39	0.39	92.52
01-Jan-2017 to 05-Dec-2017	**69,116.16**	**69,116.16**	**69,116.16**	**297.69**	**297.69**	**297.69**	**68,818.47**
Total	**408,448.77**	**408,448.77**	**408,448.77**	**1,609.36**	**1,609.36**	**1,609.36**	**406,839.41**

* For the period from 01-Dec-2017 to 05-Dec-2017, the Power Grid Company issued the confirmation for $EG_{export,y}$ and $EG_{import,y}$ of the project based on the meter reading of Meter M1.

The baseline emission during this monitoring period calculated as following:

$BE_y = EG_y \times EF_y$

Table 4. Baseline emissions

Period	EG_y (MWh)	EF_y (tCO_2e/MWh)	BE_y (tCO_2e)
01-Jan-2013 to 31-Dec-2013	95,261.24	0.9234	87,964
01-Jan-2014 to 31-Dec-2014	91,306.25	0.9234	84,312
01-Jan-2015 to 31-Dec-2015	71,824.85	0.9234	66,323
01-Jan-2016 to 31-Dec-2016	79,628.60	0.9234	73,529
01-Jan-2017 to 05-Dec-2017	68,818.47	0.9234	63,546
01-Jan-2013 to 05-Dec-2017	406,839.41	/	375,674

5.2 Project Emissions

Project emission (PE_y) is 0 tCO_2e.

5.3 Leakage

Project leakage is 0 tCO_2e.

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
01-Jan-2013 to 31-Dec-2013	87,964	0	0	87,964
01-Jan-2014 to 31-Dec-2014	84,312	0	0	84,312
01-Jan-2015 to 31-Dec-2015	66,323	0	0	66,323
01-Jan-2016 to 31-Dec-2016	73,529	0	0	73,529
01-Jan-2017 to 05-Dec-2017	63,546	0	0	63,546
01-Jan-2013 to 05-Dec-2017	375,674	0	0	375,674

Comparison of actual emission reduction and the estimation in registered PDD

Year	Estimated ER in registered PDD (tCO_{2e})	Emission reduction achieved (tCO_{2e})	Difference
01-Jan-2013 to 05-Dec-2017	485,881	375,674	-22.68%

The estimated annual emission reductions are 98,526 tCO_2e as in registered PDD. The monitoring period covers 1,800 days, 98,526 tCO_2e *1,800days/365days=485,881 tCO_2e, and the emission reduction achieved by the project in this monitoring period is 375,674 tCO_2e, less than the estimated emission reduction, which was mainly resulted by the erratically fluctuation of wind resources in the project area.

APPENDIX I: <EVIDENCE OF ACHIVED SDGS>

SDG Target 7.2, SDG Indicator 7.2.1 Renewable energy share in the total final energy consumption

period	Amount of renewable energy (MWh)	Data source
06- Dec -2007-11-Jan-2009	110,694.6026	Verification report and certification statement
12-Jan-2009-31-Dec-2009	95,904.6968	https://cdm.unfccc.int/Projects/DB/TUEV-SUED1218638332.96/view
01-Jan-2010-30- Nov-2010	98,336.7440	
01-Dec-2010 – 31-Dec-2011	117,763.0168	
01-Jan-2012-31-Dec-2012	92,778.4088	
01-Jan-2013-05-Dec-2017	406,839.41	Section 5 of this report
Cumulative amount since operation start	**922,316.879**	/

SDG Target 13.0, SDG Indicator Tonnes of greenhouse gas emissions avoided or removed

period	Amount of emission reductions (tCO_2e)	Data source
06- Dec -2007-11-Jan-2009 (VER)	102,215	Verification report and certification statement
12-Jan-2009-31-Dec-2009 (CER)	88,558	https://cdm.unfccc.int/Projects/DB/TUEV-SUED1218638332.96/view
01-Jan-2010-30- Nov-2010 (CER)	90,804	
01-Dec-2010–31-Dec-2011 (CER)	108,742	
01-Jan-2012-31-Dec-2012 (CER)	85,671	
01-Jan-2013-05-Dec-2017 (VER)	375,674	Section 5 of this report
Cumulative amount since operation start	**851,664**	/